Olympiad Inequalities

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Disclaimer

These notes are a compact compilation of inequalities encountered during Olympiad preparation.They are intended for quick revision and references for,what I believe to be the more useful and common inequalities to know for olympiad problem-solving, not as a full learning resource. For proofs and additional cool inequalities, do check out

`https://artofproblemsolving.com/wiki/index.php/Inequality#List_of_Theorems`

Arithmetic-Geometric Mean Inequality

For $a_1, a_2, \ldots, a_n \geq 0$,

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \cdots a_n}.$$

Equality iff

$$a_1 = a_2 = \cdots = a_n.$$

Weighted AM–GM Inequality

For $a_1, a_2, \ldots, a_n \geq 0$ and weights $\lambda_1, \lambda_2, \ldots, \lambda_n > 0$,

$$\frac{\lambda_1 a_1 + \lambda_2 a_2 + \cdots + \lambda_n a_n}{\lambda_1 + \lambda_2 + \cdots + \lambda_n} \geq \left(a_1^{\lambda_1} a_2^{\lambda_2} \cdots a_n^{\lambda_n}\right)^{1/(\lambda_1 + \lambda_2 + \cdots + \lambda_n)}.$$

Equality iff

$$a_1 = a_2 = \cdots = a_n.$$

Power Mean Inequality

For $a_1, a_2, \ldots, a_n > 0$ and $k_1 > k_2$,

$$\left(\frac{1}{n}\sum_{i=1}^{n} a_i^{k_1}\right)^{1/k_1} \geq \left(\frac{1}{n}\sum_{i=1}^{n} a_i^{k_2}\right)^{1/k_2}.$$

Equality iff

$$a_1 = a_2 = \cdots = a_n.$$

Cauchy-Schwarz Inequality

For real a_i, b_i,

$$\left(\sum_{i=1}^{n} a_i b_i\right)^2 \le \left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right).$$

Equality iff one vector is zero or

$$a_i = \lambda b_i \quad (i = 1, 2, \ldots, n)$$

for some constant λ.

Titu's Lemma

For real a_i and $b_i > 0$,

$$\frac{a_1^2}{b_1} + \frac{a_2^2}{b_2} + \cdots + \frac{a_n^2}{b_n} \ge \frac{(a_1 + a_2 + \cdots + a_n)^2}{b_1 + b_2 + \cdots + b_n}.$$

Equality iff

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \cdots = \frac{a_n}{b_n}.$$

Chebyshev's Inequality

If

$$a_1 \le a_2 \le \cdots \le a_n,$$
$$b_1 \le b_2 \le \cdots \le b_n,$$

then

$$\frac{1}{n}\sum_{i=1}^{n} a_i b_i \ge \left(\frac{1}{n}\sum_{i=1}^{n} a_i\right)\left(\frac{1}{n}\sum_{i=1}^{n} b_i\right).$$

If the orders are opposite, the inequality reverses. Equality iff one sequence is constant.

Rearrangement Inequality

If

$$a_1 \le a_2 \le \cdots \le a_n, \qquad b_1 \le b_2 \le \cdots \le b_n,$$

then for any permutation π,

$$\sum_{i=1}^{n} a_i b_{n+1-i} \le \sum_{i=1}^{n} a_i b_{\pi(i)} \le \sum_{i=1}^{n} a_i b_i.$$

If both sequences are strictly increasing, right equality iff $\pi(i) = i$; left equality iff $\pi(i) = n + 1 - i$.

Holder's Inequality

For $p, q > 1$ with $\frac{1}{p} + \frac{1}{q} = 1$,

$$\sum_{i=1}^{n} |a_i b_i| \le \left(\sum_{i=1}^{n} |a_i|^p\right)^{1/p}\left(\sum_{i=1}^{n} |b_i|^q\right)^{1/q}.$$

Equality iff one vector is zero or

$$|a_i|^p = \lambda |b_i|^q \quad (i = 1, 2, \ldots, n)$$

for some $\lambda \geq 0$.

Minkowski's Inequality

For $p > 1$,

$$\left(\sum_{i=1}^{n} |a_i + b_i|^p \right)^{1/p} \leq \left(\sum_{i=1}^{n} |a_i|^p \right)^{1/p} + \left(\sum_{i=1}^{n} |b_i|^p \right)^{1/p}.$$

Equality iff one vector is zero or

$$a_i = \lambda b_i \quad (i = 1, 2, \ldots, n)$$

for some $\lambda \geq 0$.

Jensen's Inequality

For convex f, x_i in its domain, $\lambda_i \geq 0$, and $\sum_{i=1}^{n} \lambda_i = 1$,

$$f\left(\sum_{i=1}^{n} \lambda_i x_i \right) \leq \sum_{i=1}^{n} \lambda_i f(x_i).$$

For concave f, the inequality reverses. If f is strictly convex, equality iff all positive-weight x_i are equal.

Bernoulli's Inequality

For $x > -1$,

$$(1 + x)^r \geq 1 + rx \quad (r \geq 1 \text{ or } r \leq 0),$$
$$(1 + x)^r \leq 1 + rx \quad (0 \leq r \leq 1).$$

For $r \neq 0, 1$, equality iff $x = 0$. For $r = 0$ or $r = 1$, equality always holds.

Schur's Inequality

For $x, y, z \geq 0$ and $r > 0$,

$$x^r(x - y)(x - z) + y^r(y - z)(y - x) + z^r(z - x)(z - y) \geq 0.$$

Equality iff

$$x = y = z,$$

or, after permutation,

$$(x, y, z) = (t, t, 0) \quad (t \geq 0).$$

For $r = 1$,

$$x^3 + y^3 + z^3 + 3xyz \geq x^2y + x^2z + y^2x + y^2z + z^2x + z^2y.$$

Muirhead's Inequality

For $x_1, x_2, \ldots, x_n \geq 0$, if

$$(a_1, a_2, \ldots, a_n) \succ (b_1, b_2, \ldots, b_n),$$

then

$$\sum_{\text{sym}} x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n} \geq \sum_{\text{sym}} x_1^{b_1} x_2^{b_2} \cdots x_n^{b_n}.$$

For positive variables, equality iff $x_1 = x_2 = \cdots = x_n$, unless the exponent tuples are the same up to permutation.

Nesbitt's Inequality

For $a, b, c > 0$,

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}.$$

Equality iff

$$a = b = c.$$

Radon's Inequality

For $a_i \geq 0$, $b_i > 0$, and $m > 0$,

$$\sum_{i=1}^{n} \frac{a_i^{m+1}}{b_i^m} \geq \frac{(a_1 + a_2 + \cdots + a_n)^{m+1}}{(b_1 + b_2 + \cdots + b_n)^m}.$$

Equality iff

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \cdots = \frac{a_n}{b_n}.$$